Supplementary Materials for "Filling in the Missing Piece: Advancing Automated Radiology Report Generation with Clinical Insights"

Chaohan Wang[1] Qi Chen[1] Yutong Xie[2] Qi Wu[1*]
[1]The University of Adelaide, Australia [2]MBZUAI, UAE
{chaohan.wang, qi.chen04, qi.wu01}@adelaide.edu.au yutong.xie678@gmail.com

1. MIMIC-1V3 Dataset Analysis

Data Split and Distribution MIMIC-CXR includes an official CheXpert [3] label file for all reports, in which the value for the positively mentioned labels is set to 1. Retaining adequate positively mentioned labels in MIMIC-1V3 is crucial for providing strong and unambiguous supervision for learning the image-text alignment. Specifically, our MIMIC-1V3 results in 119,395 training, 936 validation, and 1,546 test samples. We strictly follow the official data partition of MIMIC-CXR to ensure that our training, validation, and test samples are sourced exclusively from the original sets. Fig. 1 (a) shows the overall distribution of positively mentioned labels in MIMIC-1V3 and MIMIC-CXR.

Compared to the original dataset, six labels in MIMIC-1V3 maintain at least 40% positive mentions. On the other hand, *Support Devices* maintains only 19.1% of original positive mentions and 17% for *Enlarged Cardiomediastinum*. We consider the impact of lacking *Support Device* to be limited, as the presence of medical devices is not systematically reported in clinical practice by radiologists [2]. An insufficient amount of *Enlarged Cardiomediastinum* remains an unresolved challenge.

The label distribution in the training, validation, and test sets is shown in Fig. 1 (b). The training and validation sets have similar label distributions, while there is a noticeable discrepancy between the label distributions in the validation and test sets.

Table 1. Word counts of the indication, findings and impression in all splits with standard deviation on our MIMIC-1V3.

Split	Indication	Findings	Impression
Train	9.89 ± 6.67	45.51 ± 21.49	12.96 ± 6.17
Validation	10.23 ± 6.69	45.85 ± 22.50	12.98 ± 6.86
Test	9.94 ± 6.74	58.20 ± 23.76	16.82 ± 13.19

Lengths of Indication, Findings, and Impression Table 1 shows the word count for three sections across all splits. The length of the indication remains consistent across splits. Both training and validation sets follow a similar distribution. In contrast, the average length of findings and impression sections on the test set are 22% and 25% longer than those on the training and validation sets, respectively.

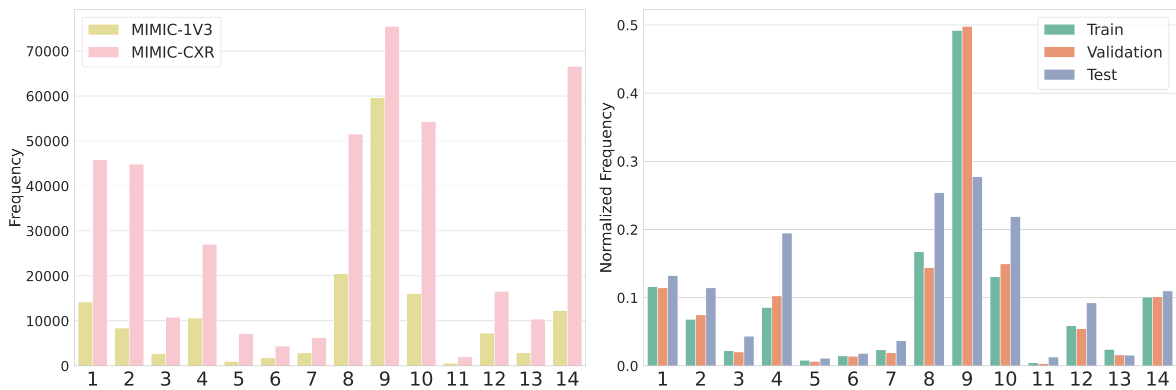
Statistical Analysis We employ Bio_ClinicalBERT [1], a domain-specific variant of BERT tailored for clinical text, to process the indication, findings, and impression sections, extracting the CLS token as aggregated information from the input sequence for calculating cosine similarities among these sections. Specifically, we compute cosine similarity of indication $\leftrightarrow$ findings , indication $\leftrightarrow$ impression, findings $\leftrightarrow$ impression.

The average of three types of similarity scores for the training, validation, test sets, and overall are shown in the Table 2. Figure 2 demonstrates the detailed distribution of three types of similarities in the entire MIMIC-1V3 dataset. Results indicate moderate similarity (mean=0.635, SD=0.120) between indication and findings, reflecting a reasonable degree of semantic overlap consistently across data splits (train, validation, and test). A higher similarity between indication and impression (mean=0.699, SD=0.100) shows strong alignment, suggesting impressions closely follow initial indications. Notably, findings and impressions exhibit very high similarity (mean=0.834, SD=0.061), indicating that impressions are almost directly derived from findings. Furthermore, a correlation of 0.71 between the indication–findings similarity and indication–impression similarity underscores a statistically robust and medically consistent relationship among these sections. To sum up, the relationship between the indication, findings, and impression is not only based on the established medical literature but also statistically robust.

2. Case studies of MIMIC-1v3

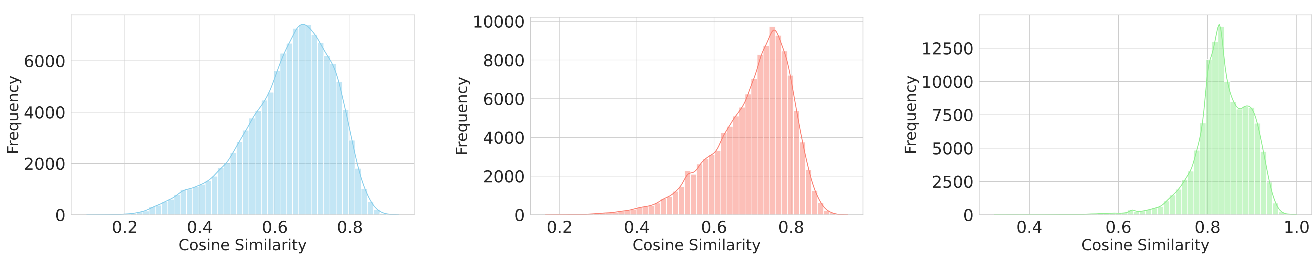
In Fig. 3, two reports from MIMIC-1V3 demonstrate how real-world data reflects the connections among the indication, findings, and impression. The pattern is clear: the indication presents a clinical question (highlighted in red),

*Corresponding author: qi.wu01@adelaide.edu.au



(a) Positively mentioned labels in MIMIC-CXR and MIMIC-1V3. (b) Normalized label distribution across splits in MIMIC-1V3.

Figure 1. Each index represents a specific label defined in CheXpert: 1: Atelectasis, 2: Cardiomegaly, 3: Consolidation, 4: Edema, 5: Enlarged Cardiomediastinum, 6: Fracture, 7: Lung Lesion, 8: Lung Opacity, 9: No Finding, 10: Pleural Effusion, 11: Pleural Other, 12: Pneumonia, 13: Pneumothorax, 14: Support Devices.



(a) The overall distribution of indication $\leftrightarrow$ findings similarity in MIMIC-1V3. (b) The overall distribution of indication $\leftrightarrow$ impression similarity in MIMIC-1V3. (c) The overall distribution of findings $\leftrightarrow$ impression similarity in MIMIC-1V3.

Figure 2. For the majority of data samples, there are strong connections among the indication, findings, and impression at the semantic level.

Table 2. Three types of cosine similarities across different data splits.

Split	IND $\leftrightarrow$ FIN	IND $\leftrightarrow$ IMP	FIN $\leftrightarrow$ IMP
train	0.635 ± 0.119	0.699 ± 0.100	0.834 ± 0.061
validation	0.673 ± 0.122	0.722 ± 0.096	0.835 ± 0.064
test	0.625 ± 0.138	0.699 ± 0.113	0.850 ± 0.065
all	0.635 ± 0.120	0.699 ± 0.100	0.834 ± 0.061

the findings section contains observations closely related to the clinical question (highlighted in brown), and the impression directly addresses the clinical question followed by primary findings (highlighted in blue).

In report (a), the indication states the patient's condition as **Hypoxia, chest pain, dyspnea**, meaning the patient has difficulty breathing. These symptoms prompt the radiologist to assess the respiratory and cardiovascular structures for abnormalities that could explain the patient's distress. **Question Infiltrate**, this specific reason for the examination directs the radiologist to scrutinize the lung parenchyma for signs of **infiltrates**, such as **patchy opacity**. **Elevated Right Hemidiaphragm** and **Prominent Azygos Vein** are the secondary signs that may relate to the primary symptoms of **dyspnea** and **hypoxia**. The impression section confirms the presence of bilateral **pneumonic infiltrates**, directly answering the clinical question posed in the indication. It also highlights other notable observations from the imaging study, ensuring that other potential issues are not overlooked and assisting the referring clinician in understanding the full scope of the patient's pulmonary condition.

In report (b), the original indication is presented as "M with c/o left lower CP with SOB and cough. ? PNA." The extensive use of abbreviations and acronyms renders the indication ambiguous and unclear. Expanding these abbreviations and acronyms to their complete forms provides a more precise context, thereby reducing model hallucinations during report generation. This issue has been addressed in the construction of MIMIC-1V3, where abbreviations have been standardized and expanded.

3. Visualization of Generated reports

To further demonstrate the superiority of our framework over other LLM-based models, we showcase how our framework effectively addresses the clinical question in the indication and generates a higher-quality report. The most relevant contents among the indication, findings, and impression are highlighted in red, while secondary findings that are worth mentioning are underlined.

As depicted in Fig. 4, the ground truth indication states, "Evaluate for aspiration or pneumonia," which requires the model to make an accurate diagnosis. The performance of RadFM is inferior; it misdiagnoses the patient with right pleural effusion and repeats most findings in the impression. Only CheXagent mentions pulmonary vasculature in the report, but it fails to identify the key finding of "low lung volumes." Since CheXagent treats the impression as the summarization of the findings and does not take the indication as the input, its generated impression, "No acute cardiopulmonary process," lacks the ability to address the clinical question and only reflects the fact that all its predicted findings are negative. Only MedVersa can identify elevated hemidiaphragm. Instead of generating descriptive and coherent sentence, MedVersa outputs only keywords and repeats the contents of findings as the impression. R2GenGPT, LLM-CXR and our framework can identify the key finding, "low volumes." However, without the indication as input, R2GenGPT tends to repeat the positive findings in the impression, which falls short of effectively addressing the clinical question. LLM-CXR exhibits severe hallucinations, mentioning previous study and neck CT. In contrast, our framework correctly predicts "without definite focal consolidation" in the findings and generates an impression that addresses the clinical question with a sound diagnosis, " No definite evidence for aspiration or pneumonia," which slightly deviates from the ground truth but matches the semantic meanings.

4. More Discussions

4.1. Effect of adding image as input for impression generation

We incorporate X-ray images as inputs for the second-stage impression generation to reduce potential error propagation between the two stages. Findings are the factual observation, or in simpler terms, the caption, of the X-ray image. Including X-ray images as input should compensate for the missing or inaccurate information in the findings. To demonstrate the quantitative effect of this mechanism, we also conduct experiments that exclude X-ray images as input for impression generation. Since the inputs are pure text, we use Lora fine-tuning to add trainable parameters to the frozen LLM. The table below shows that replacing generated findings with ground-truth findings can improve the quality of impression generation for both methods. However, the magnitude of the improvement for different methods varies.

We calculate the relative improvements after using ground truth findings as input, and for easy comparison, we calculate the mean of five clinical efficacy metrics. As shown in Table 3, without image features, ground truth findings can increase the B-4 score of the impression from 0.0383 to 0.1428, an improvement of 0.1045. This represents a 273% increase relative to the original score of 0.0383. The increases in CIDEr score and average clinical efficacy (CE) score of the impression relative to their original scores are 228% and 68%, respectively. When including images as input, ground truth findings increase the B-4 score of the impression from 0.0684 to 0.2074, an improvement of 0.1426. This represents a 220% increase relative to the original score of 0.0684. The increases in CIDEr score and average CE score of the impression relative to their original scores are 121% and 55%, respectively. These results prove that including images as input for impression generation can increase the robustness of the model, and image features can compensate for the missing or inaccurate information in the findings.

4.2. Can other LLM-based models leverage the indication?

We add the indication as input for other models and demonstrate the evaluation results in the Table 4. Our approach still outperforms these LLM-based methods.

For R2GenGPT, we add the indication to its original prompt and retrain the model. For the findings section, we observe reduced scores for all evaluation metrics except the RE-NLI, which increases from 0.2363 to 0.2398. For the impression section, adding indication as input can improve the model's performances on Bert-Score (0.3277 $\rightarrow$ 0.3678), RE-EM (0.1539 $\rightarrow$ 0.2010), and Rad-Complete (0.0869 $\rightarrow$ 0.1101), while negatively impacting the performance on B-4 (0.0532 $\rightarrow$ 0.0441), CIDEr (0.4530 $\rightarrow$ 0.2895), F1-all (0.3875 $\rightarrow$ 0.3543), and RE-NLI (0.1657 $\rightarrow$ 0.1397). Directly adding the indication without crafting R2GenGPT's prompt increased the complexity and provided no explicit instructions for the model on how to utilize the extra information, which could explain the performance drops.

For RadFM, CheXagent, LLM-CXR, and MedVersa, we incorporate the indication into their input prompts and evaluate performance on the MIMIC-IV3 test set. The inclusion of the indication improves RadFM's performance in both findings and impression generation, with consistent gains observed across all evaluation metrics. This suggests that

Study_ID: 52093225
indication: Hypoxia, chest pain, dyspnea, question infiltrate.
findings: There is patchy opacity with air bronchograms at both lung bases, consistent with a pneumonic infiltrate. The differential diagnosis could include aspiration, but this is considered less likely. There do appear to be background increased interstitial markings which could be related to either acute or chronic lung disease. Cardiomediastinal silhouette is slightly prominent, but likely accentuated by **low lung volumes**. The **right hemidiaphragm** is **elevated**. Mild prominence of the **azygos vein** is likely also accentuated by low lung volumes.
impression: Findings concerning for bilateral pneumonic infiltrates. Also diffusely increased interstitial markings, which may indicate a background acute or chronic process.

(a)

Study_ID: 54495391
indication: M with complaints of left lower chest pain with shortness of breath and cough.? pneumonia
findings: Left chest wall single lead pacing device is again seen. **Low lung volumes** are noted. Increased interstitial markings are noted in the lungs with a basilar predominance which are compatible with a chronic interstitial abnormality. There is no superimposed acute consolidation or effusion. The cardiomediastinal silhouette is stable. No acute osseous abnormalities. Hypertrophic changes are seen the spine.
impression: Findings compatible with patient's underlying fibrosis without definite superimposed acute cardiopulmonary process.

(b)

Figure 3. Two representative examples from MIMIC-1V3. The highlighted parts demonstrate the interconnections among the *indication*, *findings*, and *impression* sections.

RadFM's extensive domain-specific pretraining enables it to effectively interpret and utilize the clinical information provided in the indication, underscoring its clinical relevance from a complementary perspective. In contrast, CheXagent exhibits a performance trade-off: incorporating the indication results in a significant decline in findings generation performance across all metrics, but leads to improved impression generation, with all metrics showing enhancement. This indicates a limited generalization ability when adapting to different input modalities. LLM-CXR demonstrates relatively low sensitivity to the indication. While a marginal improvement in natural language generation (NLG) scores is observed for the impression section, a slight decrease occurs across all metrics in the findings section. For MedVersa, adding the indication consistently improves findings generation performance, with positive gains observed across all evaluation metrics.

References

[1] E. Alsentzer, J. R. Murphy, W. Boag, W.-H. Weng, D. Jin, T. Naumann, and M. McDermott. Publicly available clinical bert embeddings. *arXiv preprint arXiv:1904.03323*, 2019. 1

[2] A. Bustos, A. Pertusa, J.-M. Salinas, and M. De La Iglesia-Vaya. Padchest: A large chest x-ray image dataset with multi-label annotated reports. *Medical image analysis*, 66:101797, 2020. 1

[3] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, et al. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 590–597, 2019. 1

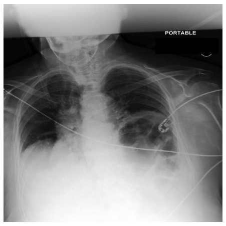
MIMIC-1V3	Ground Truth	R2GenGPT	RLPG (Ours)
Study_ID: 52526911 **Indication**: -year-old female with altered mental status and history of seizures. Evaluate for aspiration or pneumonia. 	**Findings**: Lung volumes are very low and marked elevation of the left greater than right hemidiaphragm is similar to prior. Bibasilar atelectasis is unchanged. Cardiomediastinal contours are stable. Pulmonary vascular markings appear normal. No focal consolidation or large pleural effusion. **Impression**: Low lung volumes without radiographic evidence of aspiration or focal consolidation.	**Findings**: There are low lung volumes with bibasilar atelectasis. The cardiomediastinal silhouette is not well assessed due to low lung volumes. There is no large pleural effusion or pneumothorax. Bony structures appear grossly unremarkable. **Impression**: There are low lung volumes with bibasilar atelectasis.	**Findings**: Lung volumes are low. The cardiac mediastinal and hilar contours appear unchanged. There is crowding of the bronchovascular markings without definite focal consolidation. no pleural effusion or pneumothorax is seen. There is bibasilar atelectasis. **Impression**: Low lung volumes with bibasilar atelectasis. No definite evidence for aspiration or pneumonia.
CheXagent	**LLM-CXR**	**MedVersa**	**RadFM**
Findings: The heart size is normal. The mediastinal and hilar contours are unremarkable. The pulmonary vasculature is normal. The lungs are clear. No pleural effusion or pneumothorax is seen. There are no acute osseous abnormalities. **Impression**: No acute cardiopulmonary process.	**Findings:** In comparison to study of __, low lung volumes accentuate the transverse diameter of the heart. Bibasilar opacifications are consistent with atelectatic changes. Substantial displacement of the lower cervical trachea to the right is consistent with the large thyroid mass as seen on the neck CT of __. **Impression**: mild bibasilar atelectasis.	**Findings**: Atelectasis, Elevated hemidiaphragm, Lung opacity, Vascular calcification. **Impression**: Atelectasis, Elevated hemidiaphragm, Lung opacity, Vascular calcification.	**Findings**:Right pleural effusion. Large mass projects over the right upper chest. **Impression**: Right pleural effusion. Large mass in the right upper chest.

Figure 4. Visualization of generated reports

Table 3. The quantitative effect of adding image as input for impression generation. Incorporating the images can mitigate error propagation and increase the robustness of our model. "gen" means the findings is generated from the first step. "gt" means ground truth findings.

Dataset	Method	Input	Output	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑	Avg-CE ↑
MIMIC-1V3	LoRA	IND+gen_FIN	IMPR	0.0383	0.1166	0.3305	0.3719	0.2070	0.1338	0.1103	0.2307
	LoRA	IND+gt_FIN	IMPR	0.1428 (273%↑)	0.3824 (228%↑)	0.5249	0.6042	0.3862	0.2308	0.1934	0.3879 (68%↑)
	Ours	IND+IMG+gen_FIN	IMPR	0.0648	0.7547	0.4581	0.4308	0.2773	0.2206	0.1727	0.3119
	Ours	IND+IMG+gt_FIN	IMPR	0.2074 (220%↑)	1.6740 (121%↑)	0.5897	0.6614	0.4892	0.3513	0.3306	0.4844 (55%↑)

Table 4. The evaluation results of adding the indication as input for other LLM-based models.

Dataset	Method	Input			Output		Findings							Impression						
		IMG	IND	FIN	FIN	IMPR	NLG		Clinical Efficacy					NLG		Clinical Efficacy				
							B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑
MIMIC-1V3	R2GenGPT	✓			✓	✓	0.1044	0.1530	0.5595	0.4476	0.3631	0.2363	0.1981	0.0532	0.4530	0.3277	0.3875	0.1539	0.1657	0.0869
	R2GenGPT	✓	✓		✓	✓	0.0654	0.1146	0.5426	0.3881	0.3498	0.2598	0.1971	0.0441	0.2895	0.3678	0.3543	0.2010	0.1397	0.1101
	RadFM	✓			✓	✓	0.0236	0.0576	0.3917	0.2060	0.2381	0.1846	0.1224	0.0164	0.5072	0.2289	0.2802	0.1344	0.1431	0.0893
	RadFM	✓	✓		✓	✓	0.0271	0.0603	0.4063	0.2422	0.2833	0.1963	0.1465	0.0289	0.5866	0.2409	0.2940	0.1411	0.1457	0.1245
	CheXagent	✓			✓		0.0540	0.0732	0.5152	0.2588	0.3484	**0.2639**	0.1740	-	-	-	-	-	-	-
	CheXagent	✓	✓		✓		0.0186	0.0431	0.4094	0.2216	0.2315	0.1694	0.1159	-	-	-	-	-	-	-
	CheXagent	✓				✓	-	-	-	-	-	-	-	0.0305	0.6147	0.3807	0.3749	0.2169	0.1753	0.1323
	CheXagent	✓	✓			✓	-	-	-	-	-	-	-	0.0342	0.6374	0.4019	0.3811	0.2460	0.2072	0.1499
	LLM-CXR	✓			✓	✓	0.0032	0.0072	0.3344	0.3028	0.1308	0.0294	0.0514	0.0208	0.2608	0.3593	0.3502	0.1803	0.0946	0.0860
	LLM-CXR	✓	✓		✓	✓	0.0031	0.0066	0.3118	0.2998	0.1178	0.0286	0.0457	0.0257	0.2852	0.3573	0.3595	0.1903	0.0887	0.0892
	MedVersa	✓			✓	✓	0.0003	0.0066	0.2492	0.3615	0.1362	0.0401	0.0581	0.0025	0.0477	0.3191	**0.4420**	0.1789	0.0113	0.0690
	MedVersa	✓	✓		✓	✓	0.0038	0.0235	0.2741	0.3954	0.1742	0.0721	0.0815	0.0024	0.0474	0.3194	0.4409	0.1791	0.0117	0.0698
	Ours	✓	✓		✓		**0.1155**	**0.2856**	**0.5630**	**0.4516**	**0.3685**	0.2430	**0.2026**	-	-	-	-	-	-	-
	Ours	✓	✓	✓		✓	-	-	-	-	-	-	-	**0.0648**	**0.7547**	**0.4581**	0.4308	**0.2773**	**0.2206**	**0.1727**